Shivam Gupta

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LinkedIn | GitHub

Education

Maharaja Agrasen Institute of Technology — Delhi
B.Tech in Information Technology — CGPA: 8.31 — *2024 – Present*

Lord Jesus Public School — Gurugram, Haryana
Class XII: 93% — Class X: 92%

Skills

Languages: Python, C++, Java, C
Industrial IoT: MQTT, Modbus RTU/TCP, REST APIs, Industrial Gateways, Sensor Integration
Robotics: ROS2, Nav2, Autonomous Navigation, Embedded Systems
Tools: Docker, Git, Gazebo, RViz, Fusion 360, Arduino IDE
Concepts: IoT Architecture, Networking, System Design, Data Acquisition, Edge Computing

Experience

Enercon Asia Group (GAED) — Remote
Industrial IoT Intern — *Jun 2026 – Present*

- Designed proof-of-concept Industrial IoT solutions for real-time monitoring using sensors, industrial gateways, and cloud-based dashboards.
- Evaluated Industrial IoT gateways supporting **Modbus RTU/TCP**, **MQTT**, and **REST APIs** for scalable device-to-cloud communication.
- Collaborated with hardware vendors to assess gateway capabilities, protocol compatibility, and deployment strategies for enterprise energy monitoring applications.
- Researched and documented end-to-end IoT architectures including data acquisition, gateway configuration, and API integration workflows.

ATOM Robotics Lab — Delhi
Admin — *Nov 2024 – Present*

- Engineered advanced **embedded and robotic systems** using **ESP32**, **ROS2**, and real-time control frameworks for high-impact competitions and lab deployments.
- Designed precision **mechanical assemblies** in **Fusion 360** for rapid prototyping, structural optimization, and high-accuracy **3D printed systems**.
- Developed intelligent solutions in **autonomous navigation**, **multi-robot coordination**, and closed-loop **control systems** for dynamic environments.

Projects

E-Yantra Robotics Competition 2025 Semi-Finalist
Multi Robot Warehouse Automation — ROS2, MQTT — *Video*

- Developed **multi robot warehouse automation system** with task allocation using the **Hungarian Method**, crate handling, and fleet coordination.
- Implemented navigation using **ROS2**, **ArUco marker localization**, and obstacle avoidance.
- Designed low latency communication using **MQTT publish subscribe architecture**.
- Built closed loop control using **PID controllers** for holonomic drive and docking.
- Qualified as **Semi Finalist** in the national level E Yantra Robotics Competition 2025.

Sleipnir — Quadruped Robot
Legged Robotics — Fusion 360, ESP32, Servo Control — *Video*

- Designed full **4-legged 2-DOF quadruped robot** CAD from scratch in **Fusion 360** for rapid prototyping and 3D printing.
- Built power electronics with a **200W / 20A buck converter** and **LM7805 voltage regulation** for stable ESP32 power delivery.
- Controlled **8× MG996R servos** via **PCA9685** over I2C, currently implementing **gait algorithms** and **inverse kinematics** for coordinated locomotion.

Atomtendo
ESP32 Gaming Console — Embedded Systems — *Github, Video*

- Designed custom **ESP32 based handheld gaming console** with optimized embedded firmware.
- Built hand soldered **10x20 WS2812B LED matrix PCB** with custom rendering logic.
- Implemented **millis() based non blocking controls** with smooth gameplay response.
- Developed game engine with collision detection, and **dynamic difficulty scaling** and vibration feedback.

RFID Smart Attendance System

ESP32, RFID, IoT *Github*

- Built secure **RFID based access control system** using **ESP32** and **MFRC522** with relay automation and IR triggered unlocking.
- Integrated **Firebase cloud logging**, Discord alerts, and local **OLED UI** for user enrollment.
- Utilized **FreeRTOS dual core task scheduling** for concurrent operations.

Achievements

3rd Place – Galactic Dodger Competition, IIT Kanpur Techfest

3rd Place – Line Follower Robot Competition, Delhi Technological University

Semi-Finalist – E-Yantra Robotics Competition 2025–26, IIT Bombay